

Finite-dimensional duality forces conjugation closure for RFD sub-semidual quotients

A partial answer to Open Problems 3 and 3' in arXiv:1511.07105

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Status. Candidate substantial partial result, likely valid, pending human review. For every discrete group, the finite-dimensional part of an arbitrary sub-semidual is automatically closed under conjugation. Consequently the source's Open Problem 3 has an affirmative answer whenever the associated quotient group C^* -algebra is residually finite-dimensional (RFD). The same gives RFD-qualified affirmative cases of Problems 3' and 2. No unrestricted answer is claimed.

1 Source questions and setup

Lau and Pham ask whether every weak*-closed translation-invariant subalgebra of $B(G)$ is conjugation-closed when G is discrete; equivalently, whether every sub-semidual of a discrete group is a subdual [1, Open Problem 3, p. 34]. They also formulate the locally compact Eberlein version as Open Problem 3' and note that an affirmative answer would resolve their Open Problem 2.

Open problem 3. Let G be a discrete group, and let B be a weak*-closed translation-invariant subalgebra of $B(G)$. Must B be conjugation-closed?

Equivalently, must a sub-semidual of a discrete group be already a subdual?

Corollary 3.15 shows that if G is (discrete and) amenable, then the answer to this question is affirmative. We suspect that this is not the case in general, but we do not have a counter-example. This problem could also be stated in the following form for locally compact groups.

Open problem 3. Let G be a locally compact group, and let B be a closed translation-invariant Eberlein subalgebra of $B(G)$. Must B be conjugation-closed?

If this is true, then combining with [1, Theorem 1.3] it would resolve Problem 2 in the affirmative.

Let $S \subseteq \widehat{G}$ be a sub-semidual in the source's sense. Thus S is Fell closed,

$$\operatorname{supp}\left(\bigoplus_{\pi \in S} \pi\right) \subseteq S, \quad \operatorname{supp}(\pi \otimes \rho) \subseteq S \quad (\pi, \rho \in S).$$

Write

$$U_S = \bigoplus_{\pi \in S} \pi, \quad I_S = \ker U_S, \quad A_S = C^*(G)/I_S.$$

Representations of G that factor through A_S are precisely those weakly contained in U_S . In particular, every irreducible representation factoring through A_S belongs to S .

2 Proof intuition

Finite-dimensional representations carry their own duals algebraically. If π has dimension d , then $\det \pi$ occurs in $\pi^{\otimes d}$ and

$$\bar{\pi} \cong \Lambda^{d-1} \pi \otimes (\det \pi)^{-1}.$$

Only the inverse determinant character appears to be missing. But the positive powers of any character form a semigroup in a compact group, and the closure of that semigroup contains its inverse. Fell closedness therefore supplies $(\det \pi)^{-1}$.

If A_S is RFD, its defining ideal is the intersection of kernels of its finite-dimensional representations. The preceding argument makes this family of kernels invariant under group inversion. Hence inversion descends to A_S , after which every representation factoring through A_S brings its conjugate with it. This upgrades a statement about individual finite-dimensional representations to all of S .

3 The finite-dimensional theorem

Lemma 1. *Let $\chi : G \rightarrow \mathbb{T}$ be a character of a discrete group. In the Fell topology on $\widehat{G}$, χ^{-1} belongs to the closure of $\{\chi^n : n \geq 1\}$.*

Proof. For a finite set $F = \{g_1, \dots, g_m\} \subseteq G$, put $z = (\chi(g_1), \dots, \chi(g_m)) \in \mathbb{T}^m$. The closure of the positive powers of z is a compact subgroup of $\mathbb{T}^m$; in particular, it contains z^{-1} . Equivalently, for every $\varepsilon > 0$ there is $n \geq 1$ such that

$$\max_{g \in F} |\chi(g)^n - \chi(g)^{-1}| < \varepsilon.$$

Ordering the pairs (F, ε) by refinement gives a net of positive powers converging pointwise on G to χ^{-1} . For one-dimensional representations of a discrete group, this is Fell convergence. $\square$

Theorem 2. *Let G be any discrete group and $S \subseteq \widehat{G}$ any sub-semidual. If $\pi \in S$ is finite-dimensional, then $\bar{\pi} \in S$.*

Proof. Put $d = \dim \pi$ and $\chi = \det \pi$. The character χ is the exterior-power summand $\Lambda^d \pi$ of $\pi^{\otimes d}$. Tensor closure of S therefore gives $\chi \in S$, and then $\chi^n \in S$ for every $n \geq 1$. Lemma 1 and Fell closedness give $\chi^{-1} \in S$.

Likewise $\Lambda^{d-1} \pi$ is a subrepresentation of $\pi^{\otimes(d-1)}$, so its irreducible support lies in S . The canonical finite-dimensional duality isomorphism

$$\bar{\pi} \cong \pi^* \cong \Lambda^{d-1} \pi \otimes \chi^{-1}$$

and one final application of tensor closure show that $\bar{\pi} \in S$. For $d = 1$, the same conclusion is already Lemma 1. $\square$

4 The RFD upgrade

Let $\vartheta : C^*(G) \rightarrow C^*(G)$ be the canonical linear involutive $*$ -anti-automorphism determined by $\vartheta(u_g) = u_{g^{-1}}$. If ρ is a unitary representation of G and $J : H_\rho \rightarrow \overline{H_\rho}$ is the canonical antiunitary, then

$$\bar{\rho}(a) = J\rho(\vartheta(a^*))J^{-1} \quad (a \in C^*(G)). \quad (1)$$

Theorem 3. *Let G be a discrete group and S a sub-semidual. If $A_S = C^*(G)/I_S$ is residually finite-dimensional, then S is a subdual. Equivalently, $B_S(G)$ is conjugation-closed.*

Proof. Let $\mathcal{F}$ be the family of finite-dimensional representations of G that factor through A_S . Decomposing any member of $\mathcal{F}$ into irreducibles and applying Theorem 2 shows that $\mathcal{F}$ is closed under $\rho \mapsto \bar{\rho}$.

Residual finite-dimensionality says exactly that

$$I_S = \bigcap_{\rho \in \mathcal{F}} \ker \rho.$$

Equation (1) and conjugation stability of $\mathcal{F}$ now give $\vartheta(I_S) = I_S$. Hence ϑ descends to a bounded $*$ -anti-automorphism of A_S .

Let $\pi \in S$. Since $I_S \subseteq \ker \pi$, equation (1) and $\vartheta(I_S) = I_S$ imply $I_S \subseteq \ker \bar{\pi}$. Thus $\bar{\pi}$ factors through A_S , or equivalently $\bar{\pi} \prec U_S$. The defining support condition for a sub-semidual then yields $\bar{\pi} \in S$. This is precisely the missing axiom for S to be a subdual. The coefficient-space formulation gives conjugation closure of $B_S(G)$. $\square$

Corollary 4. *Let G be locally compact and let $B = B_S^c(G) = B_S(G_d) \cap C(G)$ be a closed translation-invariant Eberlein subalgebra as in Theorem 3.11 of the source. If $C_S^*(G_d)$ is RFD, then B is conjugation-closed. If moreover B separates points, then $A(G) \subseteq B$.*

Proof. Theorem 3, applied to the discrete group G_d , makes S a subdual. Conjugation closure survives intersection with $C(G)$. In the point-separating case S is thick, and Theorem 7.1 of the source gives $B_r(G) \subseteq B$; hence $A(G) \subseteq B$. $\square$

In particular, Open Problems 3 and 3' are affirmative for all associated RFD quotients, and Problem 2 is affirmative in the RFD-associated subclass. The conclusion includes all finite-dimensional and subhomogeneous quotients, and more generally every quotient separated by finite-dimensional representations.

5 Why the apparent antipode proof is not a full solution

Lemma 3.13 of the source descends the group comultiplication to A_S , and its Lemma 3.14 observes that for discrete G the resulting object satisfies the compact-quantum-group cancellation laws. It is tempting to invoke a bounded unitary antipode R with $R(q(u_g)) = q(u_{g^{-1}})$ and conclude immediately.

That step is exactly the unresolved point for an arbitrary exotic completion. On the dense Hopf $*$ -algebra the algebraic coinverse exists, while Woronowicz's bounded polar-decomposition result is stated for the faithful-Haar form [2, Theorem 1.6]. In the present cocommutative quotient, a bounded $*$ -anti-automorphism with the displayed generator formula exists if and only if $\vartheta(I_S) = I_S$; by (1), this is equivalent to the desired conjugation closure. Invoking its descent without a separate norm argument is therefore circular. The RFD hypothesis in Theorem 3 supplies that missing norm argument by a separating family of finite-dimensional quotients.

6 Upgrade attempts, scope, and novelty

The initial full-solution attempt used the unitary antipode of the quotient compact quantum group. A deeper audit isolated the completion issue above and prevented a false promotion. A second route through determinant recurrence proved Theorem 2; the RFD kernel-intersection argument then upgraded it to Theorem 3. Attempts to obtain a counterexample from one-sided tensor powers of a finite-dimensional representation collapse by Theorem 2. A possible infinite-dimensional route through one-sided tensor families (for example holomorphic versus antiholomorphic representations) remains obstructed by the need to control weak containment after passing to the underlying discrete group.

The result does not prove that every A_S is RFD, nor that inversion is bounded on a general exotic completion. Thus it does not settle any of the three source problems without the stated RFD qualification.

A bounded primary-source search used the exact phrases “sub-semidual” and “Must B be conjugation-closed?”, the source title and authors, and the related terms “exotic group C^* -algebra”, “comultiplication”, “antipode”, and “inversion”. It covered arXiv:1511.07105, arXiv:1211.4982, and primary papers on exotic group completions, but found no later paper explicitly resolving Open Problem 3 and no explicit statement of the RFD theorem above. Novelty confidence is moderate; correctness confidence is high. Human review should focus on the Fell-limit argument for determinant characters and the RFD kernel equality.

References

- [1] A. T.-M. Lau and H. L. Pham, *Characterisations of Fourier and Fourier–Stieltjes algebras on locally compact groups*, arXiv:1511.07105.
- [2] S. L. Woronowicz, *Compact quantum groups*, in *Symétries quantiques* (Les Houches, 1995), North-Holland, 1998, pp. 845–884.
- [3] S. Kaliszewski, M. B. Landstad, and J. Quigg, *Exotic group C^* -algebras in noncommutative duality*, New York J. Math. 19 (2013), 689–711; arXiv:1211.4982.